Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/14/2026
PI Name:	Settles, Andrew Mark Ph.D.		
Project Title:	Genetic Mitigation of Spaceflight Stressors on Production of Chlamydomonas		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
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Flight Program:			
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COI Name (Institution):	Sweet, Philip Ph.D. (NASA Ames Research Center)		
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	Bioregenerative Life Support Systems (BLiSS) require diverse crops, including plants, mushrooms, and algae, that are adapted to spaceflight stressors. Chlamydomonas reinhardtii (C. reinhardtii) is a microalga that uses carbon dioxide through photosynthesis or metabolizes acetate to produce feedstocks for food, fuel, or manufacturing. The Space Algae-1 (SA-1) experiment on the International Space Station (ISS) showed that multi-generation cultivation of C. reinhardtii in space increased biomass production relative to ground controls. SA-1 revealed that space radiation increased DNA damage rates during culture, identifying a risk for genetic drift during multi-generational production. Genetic selection analysis found mutations in DNA repair and chlorophyll accumulation proteins were selected in space cultivation. Our central hypothesis is that these differentially selected genes reveal pathways required for adaptation to space radiation and microgravity. This ground study seeks to identify mitigations to reduce the impacts of radiation and		

Task Description:	microgravity to identify space-adapted strains based on ground analogs for these space stressors. Genetic variation for a subset of genes selected on the ISS during the SA-1 experiment will be engineered to test the hypothesis that these genes confer fitness effects for spaceflight stressors. Objective 1 will test the interaction of light-regulated DNA repair pathways with mutation frequencies induced by simulated Galactic Cosmic Radiation (GCRsim) at the NASA Space Radiation Laboratory (NSRL). Wild-type and genetic variants for selected DNA repair pathway genes from the SA-1 experiment will be exposed to radiation with either continuous lighting or light/dark growth cycles to identify specific factors that mitigate mutation rates. We hypothesize that allowing for light/dark cycling and incorporating space-selected mutations in DNA repair proteins will lower the rate of mutations after GCRsim exposure. Objective 2 will investigate the interaction between chlorophyll accumulation and simulated microgravity. In SA-1, microgravity induced differences of cell distributions in liquid culture caused differential light exposure, and genetic selection for variation in chlorophyll accumulation genes was detected. Variants for space-selected chlorophyll accumulation genes will be cultured in simulated microgravity to determine growth rates and photosynthetic parameters. We hypothesize that strains expressing space-selected mutations in chlorophyll accumulation genes will affect growth rate or total biomass relative to wild-type strains in simulated microgravity. Objective 3 will test whether broad abiotic stress tolerance predicts tolerance for GCRsim or microgravity using the recently discovered extremophile, Chlamydomonas pacifica (C. pacifica). We hypothesize that C. pacifica will show more robust growth in simulated microgravity and greater tolerance to GCRsim exposure compared to C. reinhardtii. This project directly addresses the Space Crops research emphasis for the ROSES-24 E.9 Space Biology solicitation. This hypothesis-driven investigation will test genetic and physical mitigations for two space-related stressors that impact algae production in space. The project addresses the 2023 BPS Decadal Survey key science question (KSQ) 1, that calls for investigating the influence of space stressors on biological mechanisms required for thriving in space. KSQ2 will be addressed by examining the mechanistic impact of genetic diversity in response to spaceflight stressors. KSQ4 will be addressed by testing specific mutations identified as differentially selected in space during a multi-generational competitive growth experiment. The project also aligns with the recommended BLiSS research campaign by studying an organism that can be directly incorporated into systems to produce vitamins, food, and biomaterials from waste streams, while revitalizing air.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• Identify space-adapted genotypes for more resilient biological life support systems and biomass production. • Better understand genetic diversity and mutation targets of algae exposed to space-related stressors. • Identify how light/dark cycling affects DNA repair for space crops exposed to space stressors.
Task Progress:	New Project for FY2026.
Bibliography Type:	Description: (Last Updated: 05/04/2026)